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Geometry controls momentum flux in the sprinkler problem

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Hydro- and aero-mechanical devices convert fluid flows into useful motions, force, and power. The operating principles can involve subtle and poorly understood physics, as epitomized by open questions about systems that aspirate flows through curving tubular arms. Since its introduction by Mach and popularization by Feynman, the so-called reverse sprinkler problem has evoked many competing theories and fluid mechanical effects that have not to date been distinguished by experiments. Here we conduct a series of experiments that directly report on the motions, torques, and flows for devices whose geometries are tailored to disambiguate the leading hypotheses. Our observations run counter to several ideas, such as those based on the total angular momentum of the fluid and others focusing on the flow and pressure distributions at the outer portions of the arms. The measurements instead reveal strong correlations between the sense of torque/rotation and the fluid momentum fluxing into the device. These results suggest an operating principle for the reverse sprinkler involving isotropic input of fluid from the far field and swirl-up in the arms that generates angular momentum, a residual portion of which is injected inside and drives rotation. The mass-to-momentum flux conversion is governed by the geometry of the curving arms. The physics learned here is fundamental to flow–structure interaction problems and may inform applications for harvesting and transforming flow energy.

flow–structure interactions | hydromechanics | energy harvesting | flow control | motion control

Significance

Many technologies transform wind or water currents into useful forces and motions of structures. Our study makes significant progress toward understanding the physics of hydromechanical devices called sprinklers that rotate by expelling or suctioning fluid. We provide comprehensive motion, torque, and flow measurements on designs that distinguish among competing ideas for the operating principle. By showing that fluid momentum flux is the dominant mechanism, our findings address a long-standing open problem in flow physics and provide useful knowledge about how these devices work and their effectiveness. The results also include specific guidelines for how to design structures that control flows to produce torque and rotation.

Flowing air and water are abundant sources of energy that can be extracted and stored or otherwise harnessed for applications, and the associated technologies involve the flow-induced forcing and motion of solid structures and mechanical components (1–3). The practical importance of such devices is complemented by their scientific role of informing on flow–structure interaction physics and the fundamental principles governing systems of fluids and solids that exchange energy and momentum (4–6). Particularly useful for these purposes have been the class of rotary machines called sprinklers that spin in response to ejecting fluid from tubular arms. Familiar from their everyday use as self-spinning lawn sprinklers and autorotating dishwasher blades, such systems have featured prominently in investigations into action–reaction principles and the conservation of angular momentum (7–9). An enduring case is the so-called reverse sprinkler problem that asks about the response of structures that draw in ambient fluid through curving, S-shaped arms. This problem has attracted considerable interest and earned recognition for exposing subtle and challenging issues that have provoked disagreements about the relevant mechanisms and more generally about the analysis of open systems subject to mass flux and flow throughput due to sources and sinks (10–23).

A common approach to the reverse sprinkler question involves reasoning about the possibility of rotation based on momentum conservation principles applied globally to the fluid–solid system. This view was the original intent of the problem as raised by Ernst Mach, who reasoned that neither fluid nor solid has significant angular momentum in steady state (7). In the surge of interest following Richard Feynman's famous account of failed experiments on the problem (8, 10, 11), studies aimed to apply conservation principles but arrived at conflicting predictions as to whether the reverse sprinkler rotates and, if so, in which direction (12–14). Experimental works that followed tended to show reverse rotations but were unable to explain the outcomes using conservation of total angular momentum (15, 17, 18). These findings raised questions about the adequacy of principle-based argumentation to infer the response, and later studies increasingly shifted focus to more nuanced hydrodynamic issues. Much attention has been given to the sink-like flow entering the outer nozzles, where the effects of suction pressure and fluid impingement are argued to counteract and exactly cancel in the inviscid limit (19–21, 23). Accounting for reverse rotation then spurred discussions about the roles of fluid viscosity and turbulence (21, 22). Another view of the problem has looked inward to the central hub of the sprinkler where the aspirated fluid exits the arms as jets that collide and induce complex flows (9, 16, 22). Proposed mechanisms include shearing

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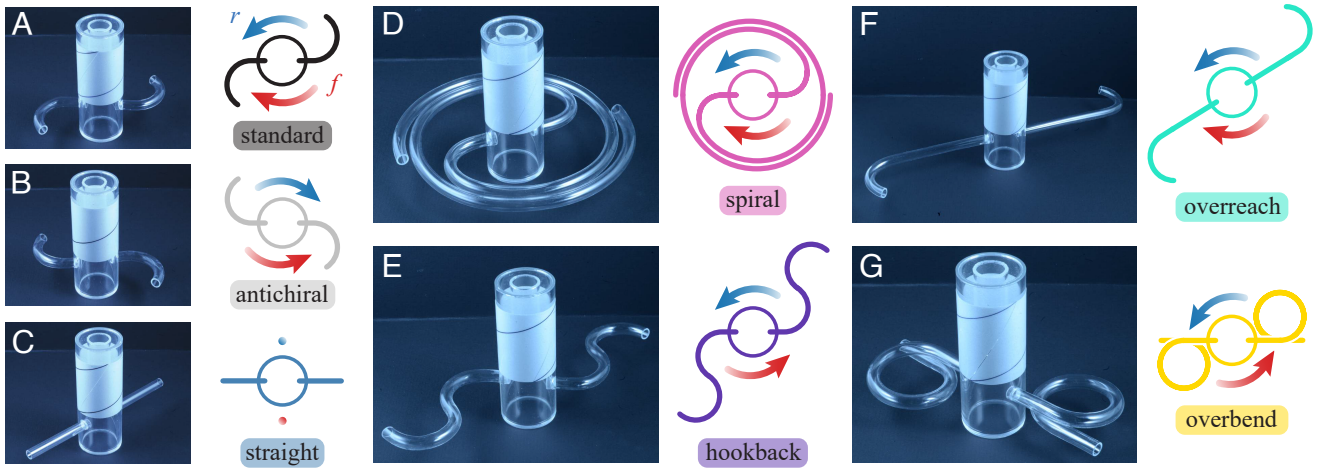


Fig. 1. Sprinkler geometries and their observed responses. Photographs of experimental devices are accompanied by schematics showing the observed senses of rotation and torque (arrows) for the forward (f, red) mode in which fluid is expelled from the curving arms and reverse (r, blue) mode in which fluid is suctioned in. The designs are grouped according to controls (*A*–*C*), those modifying the fluidic angular momentum (*D* and *E*), and those modifying the location and orientation of the outer orifice of the arms (*F* and *G*).

action on the inside walls of the hub by vortices and flux of angular momentum into the hub due to asymmetries in the jet velocity profiles. The latter view was recently supported by experimentally measured rotations that could be explained by flow measurements and drag modeling (9).

From reviewing the literature, it remains unclear if the various proposed mechanisms are physically distinct or instead differ as matters of emphasis (e.g. flow versus force). Further, there is a general deficit of experimental data that could distinguish the hypotheses, and there are to date no torque measurements that would directly report on the mechanisms. Here we aim to disambiguate the leading theories by interrogating different geometries of sprinklers for which we provide a battery of motion, torque, and flow measurements. We organize our work around three main classes of theories identified in the literature, namely those based on total angular momentum of the fluid, the imbalance of flux and pressure at the outer orifice, and the flux of angular momentum into the central hub. Our strategy is to devise and experimentally test geometries that are expected to either amplify or nullify the main effect emphasized by each hypothesis. The measurements therefore speak to the power of each theory to predict the response and explain the observations. This research program is intended to fill the void of experimental results in the literature, break the impasse regarding the operating mechanism, and provide critical information for future models and simulations.

Survey of Geometries and Their Observed Behaviors

We first introduce the systems to be studied and qualitatively describe the observed motion and torque responses. The basic units are displayed in Fig. 1 with photographs of the manufactured devices and schematics of their cross-sections. The design specifications allow for measurements of their free rotation, static torque, and accompanying flow fields using experimental apparatuses to be described in detail. The devices operate submerged under water with flow either issuing out from the arms in the forward (f) mode or suctioned into the arms in reverse (r). Each consists of a cylindrical hub that is common across all designs and which is open at the top to allow flow input or withdrawal. Tubular arms enter through the hub wall, and their centerline shape distinguishes the different designs which in all cases have two-fold rotational symmetry. Each arm has an outer opening or orifice that issues out flow in the forward mode and aspirates fluid in the reverse mode, and an inner orifice is the intake in the forward mode and exhaust in the reverse.

The specific geometries are designed to distinguish the leading ideas from the literature. The first group shown in Fig. 1 *A*–*C* includes the standard S-shaped type against which others will be compared. The antichiral geometry is its mirror image and serves as a test of the manufacturing methods, and the straight arm design is a further control. The second group shown in Fig. 1 *D* and *E* tests the hypothesis involving total angular momentum of the fluid, which in the reverse mode is expected to dominantly derive from the flows inside the arms. The spiral sprinkler ought to greatly increase this fluidic momentum due to the significant azimuthal extension of the arms, whereas the hookback design has a momentum canceling effect due to the addition of the counterdirected outer bend. The third group of Fig. 1 *F* and *G* tests the role of forces developed at the outer orifice and its connection to the elbow or bend. The overreach design ought to enhance such effects by providing a longer lever arm for the relevant forces, whereas the overbend geometry ought to nullify or weaken such effects since the line of action of the forces projects through the center of the device.

Across all geometries in both modes, it will be shown that the observed rotational motion of the free structure is in the same direction as the torque generated when it is held static. The rotation/torque directions are shown as curved arrows on the schematics of Fig. 1 for the forward (f, red) and reverse (r, blue) modes. The forward mode has few surprises, and most cases are readily understood as consistent with the rule that the structure rotates in the opposite direction to the flow issuing from the outer orifices. Correspondingly, the straight arm design rotates little and without clear direction. The overbend represents a case not easily rationalized a priori, and it is found to rotate in the forward mode oppositely with respect to the standard design.

For the reverse mode, the controls play out as expected but otherwise the results defy expectations based on the hypotheses motivating the designs. The spiral and hookback geometries rotate in the same direction as the standard and produce similar

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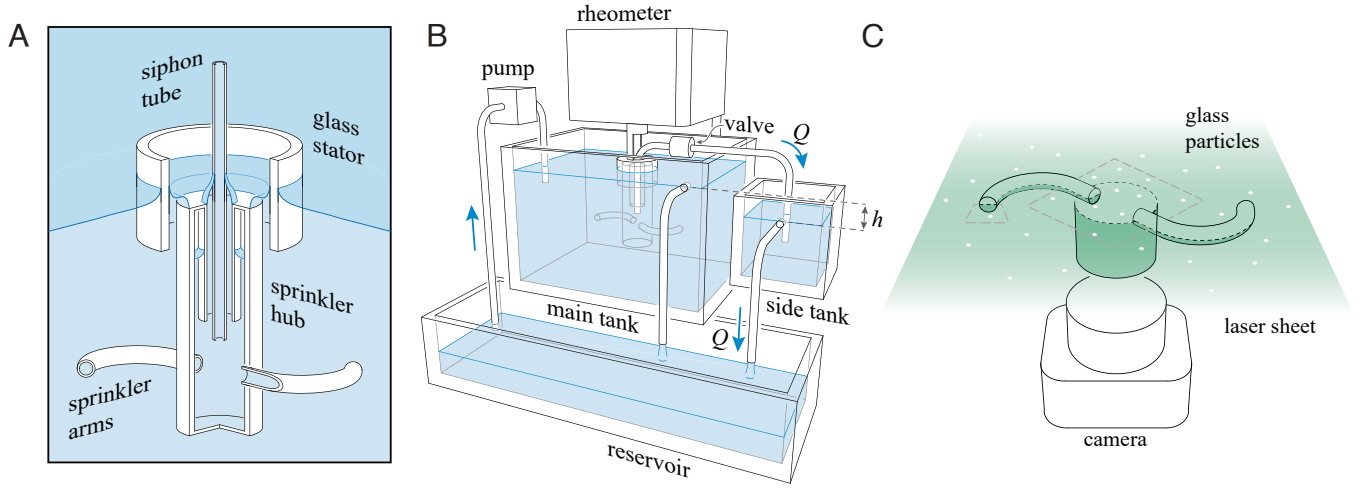


Fig. 2. Experimental setups. (*A*) Cut-away schematic of the sprinkler mounted in a surface tension bearing, allowing for free rotation without the friction incurred by solid–solid contacts. (*B*) Flow control apparatus and sprinkler mounted on the rheometer for torque measurement. The flow throughput Q is set by the relative height h of the free surfaces. (*C*) Flow velocimetry using a laser sheet through the plane of the arms and a camera capturing high-speed movies of particle motions.

torques, outcomes that contradict predictions based on reaction to the total angular momentum of the fluid. The overreach and overbend geometries in reverse mode also rotate in the same direction and produce similar torques as the standard, defying expectations based on forces at the outer orifice. This point is reinforced by the hookback, which has its orifices oppositely directed compared to the standard and yet its motion and torque are in the same direction. This survey brings the focus to the idea of angular momentum flux, which is not easily discerned for any geometry in the reverse mode and therefore motivates a detailed analysis of the flows. The sections that follow provide the methods and measurements that substantiate these conclusions.

Experimental Setups and Procedures

We conduct three types of experiments for each geometry with the goals being to characterize the rotational motion of the free structure, the steady-state torque on the static structure, and the flows generated near the arm orifices for the static case. Each involves dedicated apparatuses into which each sprinkler is installed, and key parts of the setups are shown in Fig. 2 *A*–*C*. More complete descriptions are given as *Materials and Methods*. All three systems involve appropriate plumbing for supplying or withdrawing fluid at a controlled and measured volumetric rate Q, and for simplicity these components are shown only in panel (*B*). A side tank and siphon provide the flow input/output for the sprinkler, which sits at the surface of a main tank whose water level is maintained by an overspill and pump system. This allows for long measurement periods so that steady states are reached and maintained.

Motion measurements of freely rotating sprinklers are enabled by mounting the sprinkler in a custom fluidic bearing, as shown in Fig. 2*A* and described in detail elsewhere (9). The bearing is a capillary trap that holds the sprinkler in place while permitting its rotation, and it exploits repulsive forces from surface tension interactions between oppositely signed menisci to loosely confine the sprinkler in an annular region between the siphon tube on the inside and a ring on the outside. These two are fixed in the lab as stators whose glass surfaces induce upward menisci, while the sprinkler acts as the rotor whose downward meniscus is produced by slight negative buoyancy controlled by an internal pocket of air. Critically, there is no solid–solid contact with the sprinkler, whose rotation is therefore dominantly resisted by fluid drag. The flow throughput Q is controlled as in Fig. 2*B*, which determines the Reynolds number $\mathrm{Re} = \rho UD/\mu = 2Q/\pi\nu D$ for the pipe flow of mean speed U in each of the two arms (inner diameter $D = 0.6$ cm). The bearing has been validated for $\mathrm{Re} \in [50, 500]$, the lower bound of which marks the onset of reverse motion and yet stronger driving is associated with increasing susceptibility to loss of centering (9). We therefore interrogate $\mathrm{Re} = 500$ for which we anticipate robust motion in the reverse mode, and we focus our efforts on conducting multiple trials for each geometry in both modes to ensure reproducibility. For each run, the angle $\theta(t)$ and thus angular velocity $\Omega = \dot{\theta}$ are measured from videos that allow motion tracking of a helical stripe taped to the hub's outer wall, i.e. the barber's pole seen in the photos of Fig. 1.

The torque measurements are performed by mounting the sprinkler on a precision rheometer, as shown in Fig. 2*B*. The apparatus directly reports the torque required to keep its spindle fixed, and the sprinkler is attached to the spindle while flow is routed through the top. As detailed in *Materials and Methods*, we determine that the noise floor of the rheometer's torque measurement to be about 1 dyn·cm, which sets the lower bound on Re that can be assessed for the reverse case. We exploit the automation of the device to sweep a broad range of $\mathrm{Re} \in [200, 1{,}500]$. The temporal resolution of the readout is 1 Hz, and our procedure involves waiting sufficiently long for steady state to be reached and then averaging over the time-series data obtained over a duration of 480 s. The static torque measured in this way is indicative of the propulsive torque generated by a freely spinning sprinkler. This correspondence may be seen as direct for the reverse case given that the speeds associated with the flow throughput are much greater than those due to rotation.

Flow velocimetry measurements involve the setup shown in Fig. 2*C* for which the sprinkler is fixed in place. The water is seeded with glass microbubbles of typical size 50 μm, and the plane through the midheight of the arms is illuminated by a laser sheet. Similar methods have been applied and validated in previous works (9, 24, 25). The particles scatter the laser light and appear as bright points in videos captured from below with

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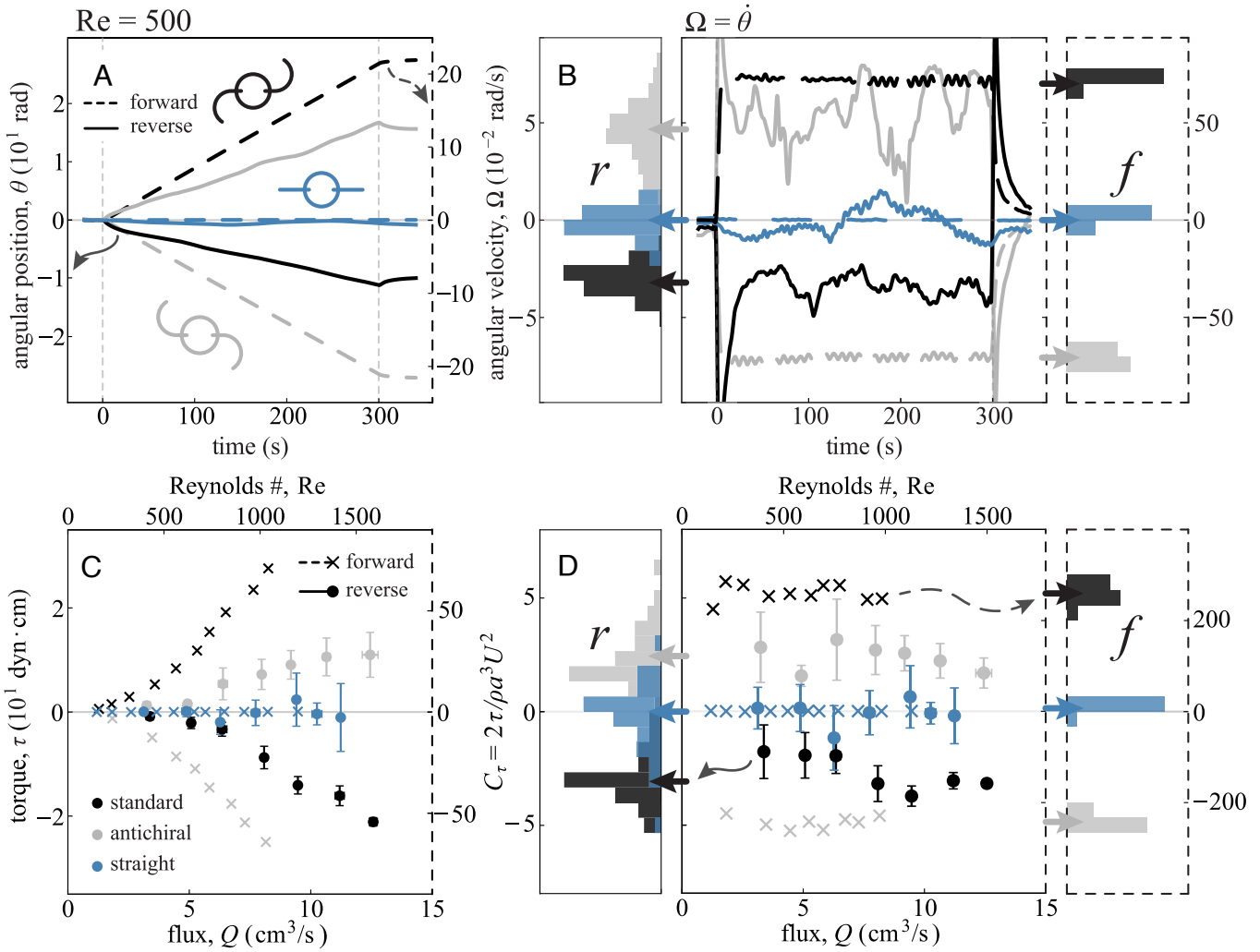


Fig. 3. Motion and torque characterizations of the standard (black curves and markers), antichiral (gray), and straight (blue) geometries. (*A*) Example trials showing measured angular position over time at Re = 500 for the forward (dashed curves) and reverse (solid) modes, which are shown on differing vertical scales as indicated by the arrows. (*B*) Corresponding angular velocity time series, where data from repeated trials fill the histograms for the forward (f, *Right*) and reverse (r, *Left*) modes. (*C*) Mean and SD of the measured steady-state torque across varying *Q* or Re for all geometries and modes. (*D*) Corresponding torque coefficient flanked by histograms for both modes.

a high-speed camera. As detailed in *Materials and Methods*, we apply particle-image velocimetry (PIV) algorithms to extract flow velocity fields with high spatial and temporal resolution (26–28), which are then averaged in time to determine the steady-state flow metrics. We interrogate the regions indicated in Fig. 2*C* near the inner and outer orifices that are critical to the exchange of momentum with the surrounding fluid and which therefore inform on the flux hypothesis. Based on previous findings that the mean flow profiles change relatively little for Re > 200 while becoming increasingly unsteady (9), here we focus our investigations on Re = 250.

The Standard Sprinkler and Control Studies

We next present motion and torque data for the standard sprinkler and its controls and use these cases to justify procedures and analyses to be applied across all geometries. The rotation responses for the forward and reverse cases of the standard, antichiral, and straight geometries are shown in Fig. 3. Panel (*A*) displays typical time-series data for the sprinkler angle $\theta(t)$ and the central plot of B shows the corresponding angular velocities $\Omega = \dot{\theta}$. Curve colors differentiate the geometries, and line styles distinguish forward (dashed) and reverse (solid) modes. The data reflect an experimental program in which the flow is initiated impulsively at time $t = 0$, maintained for 5 min to ensure steady state, and then abruptly stopped. The speeds in reverse are about an order slower than in forward, so the plots of A and B are shown with differing vertical scales as labeled on the left axis for reverse case and the right for forward. The quiescent periods before and after verify that the motions derive from flow throughput and not effects such as ambient currents in the tank caused by the pump-overspill system. The motions are directional for the standard and antichiral geometries in each mode and oppositely so between the two designs, whereas the movements are weak and undirected for the straight geometry in both modes. The latter supports the interpretation that the surface tension bearing by itself does not impart significant propulsion, and its resistance is discussed as *Materials and Methods*.

The mirror-symmetric responses of the standard and its antichiral counterpart for both modes indicate that the motions derive from the chiral arrangement of the arms. For both geometries, the direction in the forward mode is consistent with

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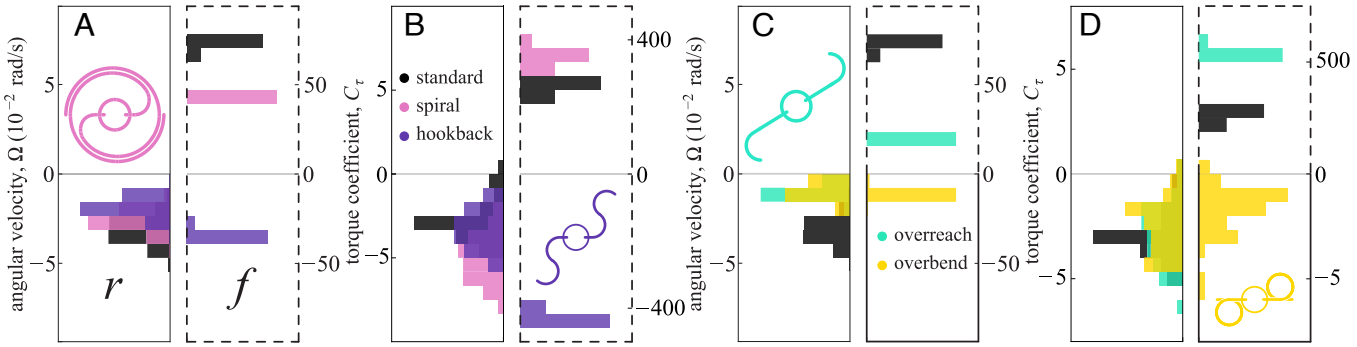


Fig. 4. Summary histograms of speed and torque data for geometries probing the leading mechanisms. (*A* and *B*) Steady-state angular velocity Ω and torque coefficient C_τ for the spiral (pink) and hookback (purple) as compared to standard (black). Each panel includes the reverse (r, *Left*) and forward (f, *Right*) mode data on different scales. (*C* and *D*) Corresponding histograms for the overreach (green) and overbend (yellow) designs.

expectations based on propulsion opposite to the fluid jets issuing from the outer orifices, and the reverse motion is oppositely directed. The forward motion is quite steady whereas the reverse mode shows considerable variations that are most apparent in the velocity data of Fig. 3*B*. A speed burst just after turn-on gives way to terminal behavior that is reached in all cases after about 30 s and which we term "steady state" despite the fluctuations. Flanking the time-series plot of B are histograms of the reverse (*Left*) and forward (*Right*) speeds from three such runs for each geometry/mode condition, and only steady-state data for $t \in [100, 300]$ s are included. The magnitudes for the standard versus antichiral designs show differences in their means of 2% for the forward mode and 35% for reverse. These values reflect unintended differences in manufacturing the sprinklers, and the higher sensitivity of the reverse mode accords with its slower speed and the presumably more subtle ways in which its mechanism depends on geometry.

The torques τ versus Q and Re for the standard, antichiral, and straight geometries are assessed in Fig. 3*C* where the color coding is maintained and the marker symbols distinguish forward (crosses) versus reverse (dots). The differing vertical scales on the left (reverse) and right (forward) axes reflect the nearly two orders difference in magnitudes between the modes, and the lower and upper axes indicate Q and Re. Each data point reflects time-averaging over an eight-minute duration and ensemble-averaging over 4 such trials, and the displayed bars correspond to SDs. These torque data recapitulate the key outcomes of the rotation speed measurements. The standard and antichiral geometries tend to have mirror-symmetric responses in both modes, and the straight arm design generates weak and undirected torque in all cases. The standard and antichiral responses are notably nonlinear in $Q \propto \mathrm{Re} \propto U$, which motivates the nondimensional torque coefficient $C_\tau = \tau / \frac{1}{2}\rho a^3 U^2$ based on the scales of the pipe-arms. The data are recast in D where the near constancy of the values indicates that a quadratic scaling with speed largely accounts for the flow dependence. Hence each geometry in each mode can be well characterized by histograms of the coefficient, as shown to the left (reverse) and right (forward). The means for the standard and antichiral sprinklers differ by 5% in the forward mode and 20% in reverse, and these values reflect the sensitivity of torque to unintended variations in construction.

Testing the Total Fluid Angular Momentum Hypothesis

Next we characterize the spiral and hookback geometries, which are designed to respectively enhance and neutralize the total angular momentum of the fluid in the reverse mode. The dominant contribution to fluid momentum is expected to come from the flow coursing through the arms, as based on the outer flows being sink-like and confined to small regions near the outer orifices and the inner flows involving vortices inside the hub but whose opposite signs and small radial distances suggest weak momentum (9). The angular momentum of the flow can therefore be associated with the azimuthal extent of the arms, from which the spiral is predicted to have substantially stronger response than the standard whereas the hookback is weaker due to its counterbend.

The measured motion and torque responses are summarized as histograms in Fig. 4 *A* and *B*, and the full sets of underlying measurements are given as *Materials and Methods* and the associated extended data figure. The main conclusion about the reverse (r, *Left* panels) mode is that, for both angular velocity and torque, the spiral and hookback designs fail to produce changes in responses commensurate with the anticipated changes in flow angular momentum relative to the standard. The *Left* panel of (*A*) shows that the reverse velocities among the three geometries are of the same sign and have similar histograms with overlapping ranges. The mean speeds for the spiral and hookback are respectively 20% and 50% lower than the standard. Similarly, the *Left* panel of (*B*) shows that the torques are of the same sign and also only weakly different among the three geometries, and the means are respectively 50% and 10% greater than the standard. The spreads in the speed and torque data are of similar order, and the control studies have shown comparably large variations. Taken together, these results indicate that the total fluid angular momentum hypothesis has at best weak power to explain the reverse sprinkler.

In the forward mode, the spiral design rotates and produces torque in the same direction as the standard, whereas the hookback rotates and torques oppositely. These signs are consistent with the rule that the forward mode is driven oppositely to its outer jet flows, which are flipped azimuthally for the hookback. The hookback therefore stands out as a sprinkler that responds in the same direction in the two modes, albeit with very different magnitudes of speed and torque. For both modes, the relative ordering among geometries is not necessarily preserved when comparing their magnitudes of velocity versus magnitudes of torque. For example, the hookback in forward mode produces more torque than both the standard and spiral but spins more slowly than either. This can be understood as the effect of hydrodynamic drag, which balances propulsive torque to determine the speed in steady state and which is expected to be relatively high for the hookback given its radially extended arms.

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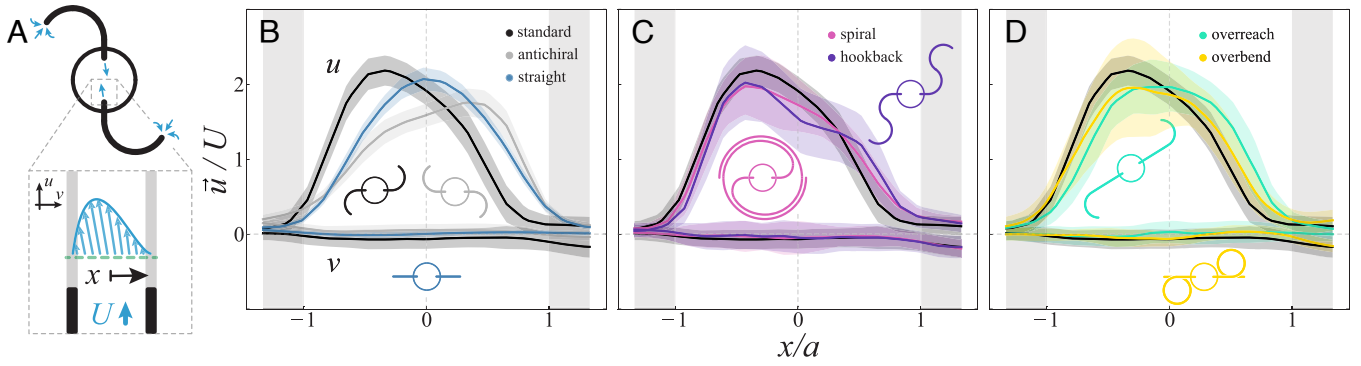


Fig. 5. Flow profiles at the inner orifices in the reverse mode. (*A*) Velocimetry measurements resolve the axial u and transverse v components along transects through the jets entering the hub at Re = 250. (*B*) Means (curves) and SDs (shaded bands) of flow components for the standard, antichiral, and straight designs. The coordinate is normalized by the inner radius a of the pipe-arms and the speeds by the mean $U = Q/2\pi a^2$. (*C* and *D*) Profiles for the spiral, hookback, overreach, and overbend designs.

Testing the Outer Orifice Hypothesis

We next take up the overreach and overbend geometries that are tailored to assess whether the forces developed at the outer portions of the arms are dominantly responsible for the dynamics of the reverse sprinkler. Previous studies have emphasized mechanisms that may lead to imbalanced pressures acting along the entrance region from the outer orifice to the bend or elbow of the arm (20, 21, 23). Hence it has been proposed that extending the arms radially outward ought to increase the reverse torques (23), and this motivates the overreach geometry with its few-fold increase in the lever arm. The overbend design aims to largely neutralize the torque developed from such forces by directing their line of action through the center of the device, which is accomplished by reorienting the relevant segment of the arms to point radially rather than azimuthally.

The angular velocity and torque measurements are summarized by the histograms of Fig. 4 *C* and *D* where data for the standard geometry is repeated for reference. The main conclusion apparent from the *Left* panel of (*D*) is that the overreach and overbend designs fail to modify the reverse torques relative to the standard in ways commensurate with the view that forces at the outer portions are the prime driving mechanism. The mean torques are within 20% of the standard for both geometries, and these differences are small compared to the variability in the measurements. Moreover, we note that the hookback design should flip the sign of the torque if such forces were dominant, and the data of Fig. 4*B* show no such effect. Hence we reject the outer orifice hypothesis as a primary explanation for the reverse sprinkler. Note that the reverse speeds shown in the *Left* panel of (*C*) for the overreach and overbend designs are less than the standard, which is consistent with the higher drag expected due to the greater lengths of the arms.

The forward dynamics respond strongly to the changes to the outer orifices in ways consistent with reaction to the emitted jet flows. The forward torques shown in the *Right* panel of Fig. 4*D* show that indeed the overreach has several-fold increase relative to the standard, and the overbend has about 100-fold decrease (note the greatly reduced scale on the vertical axis). The forward velocities displayed in the *Left* panel of (*C*) show a commensurate slow-down of the overbend (note again the axis scale), while the overreach slows relative to the standard despite its higher torque. This suggests that the benefit to the propulsive torque provided by the longer arms is outweighed by the increase in drag.

Testing the Angular Momentum Flux Hypothesis

We turn our attention to the inner orifices as sites where fluid angular momentum may be injected into the hub. The flows entering the hub take the form of inward jets whose velocity profiles may be subtly asymmetric in ways that produce a flux of angular momentum (9). The flows are not easily predicted from the arm geometry and hence we conduct velocimetry measurements for all 7 sprinkler designs. As diagrammed in Fig. 5*A*, these experiments yield the axial u and transverse v components of velocity within a plane through the middle of an orifice and across the position x along a transect just inside the hub. The flux can be expressed as an integral over the velocity profile (see the section that follows) that may be nonzero if $u(x)$ is skewed with a maximum away from the center or if $v(x)$ is dominantly negative or positive. The features of the jet flow profile consistent with the observed direction of rotation/torque for the reverse mode of the standard sprinkler are highlighted in the zoomed-in window of panel (*A*). The axial flow is skewed leftward as defined based on the bend of the arm, and the transverse flow is biased leftward such that $v < 0$. The flux hypothesis posits that the jets inject angular momentum of one sign and the sprinkler responds with oppositely signed velocity and torque.

The flow profiles for the control group are shown in Fig. 5*B*, where the u and v components are scaled relative to the mean axial speed U and displayed across the normalized transect coordinate x/a. Each case is characterized by its time-averaged profile (colored curve) and SD (similarly shaded band) as determined by analysis of high-speed video. The flow is dominantly axial with u about two orders greater than v. The straight-armed geometry has parabolic-like axial flow with uniformly small transverse component as expected for straight pipes (29). In contrast, the standard design has skewed u and negative v, both of which contribute to an angular momentum flux signed consistently with the observed sprinkler motion/torque. Moreover, the antichiral design has the opposite sense of skewness and oppositely signed transverse flow consistent with its opposite motion and torque. More surprising are the results of Fig. 5 *C* and *D* for the spiral, hookback, overreach, and overbend geometries, all of which show leftward skewness of u and leftward (negative) bias of v. These results are concordant with these geometries showing reverse motions and torques similarly signed as the standard case. The overreach design has a less skewed axial flow profile than the

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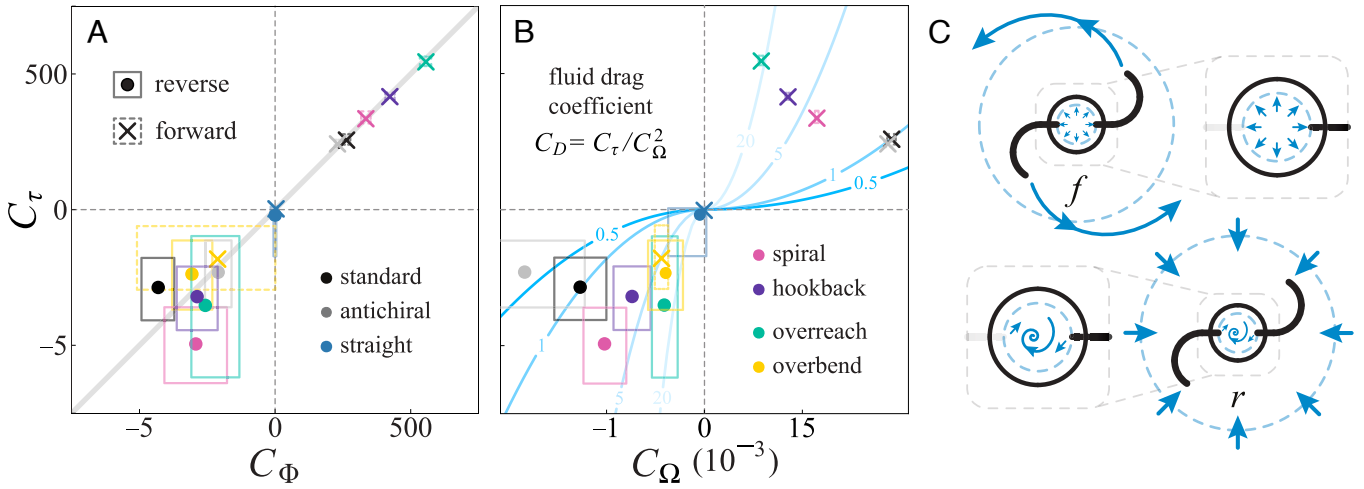


Fig. 6. Angular momentum flux explains torque production across all designs in both modes. (*A*) Dimensionless torque versus angular momentum flux. The relation $C_\tau = C_\phi$ (gray line) explains the trends and scales for the forward (crosses) and reverse (dots) modes of all 7 geometries. Boxes represent SDs, and the scales differ for the two modes. (*B*) Dimensionless torque versus speed. The curves are labeled with values for the drag coefficient normalized by that of the standard design. (*C*) A common operating principle for both modes involves the transformation of isotropic input flows into chiral output flows carrying angular momentum opposite to the sprinkler rotation/torque.

standard, and this is consistent with the distorted flow at the elbow being somewhat symmetrized during the longer transit to the hub (*Materials and Methods*). Together, these outcomes raise the flux hypothesis above others as a promising explanation for the reverse sprinkler.

Sprinklers as Mass-to-Momentum Flux Transformers

We next ask if the angular momentum flux hypothesis can quantitatively account for the propulsive torques. We consider a theory in which the torque is due entirely to the fluidic angular momentum fluxing through the inner orifices in the reverse mode and through the outer orifices in the forward mode. The latter case is motivated by the intuition that the forward sprinkler acts as a rotational analog to a rocket, the classical model of which involves a propulsive force of equal magnitude and opposite direction to the flux of linear momentum of the expelled propellant (30). The relevant angular momentum flux can be expressed as the integral of the flow velocity field over the disk-shaped domain: $\Phi = \iint_{\mathrm{IO,OO}} (\rho \vec{r} \times \vec{u}) \vec{u} \cdot d\hat{S}$ where $\vec{u} = (u, v)$, $\vec{r}$ is the position from the axis of rotation, and the domain corresponds to the inner orifice (IO) in the reverse mode and outer orifice (OO) in forward. The critical term in parentheses is the density of fluidic angular momentum, and the flux-based torque associated with the two orifices is $\tau_\Phi = -2\Phi$. This model is understood to be highly idealized since all other contributing effects are neglected, but nonetheless comparisons to the measured torques can inform on whether the flux is an order-one participant in the dynamics. The formulation relates the flux-based torque to the 2D flow field across the face of the orifice, which can be determined from the measured 1D transect data of Fig. 5 by the interpolation procedure provided as *Materials and Methods*.

The results of the model are compared to the experimental measurements in Fig. 6*A*, where the torques are cast as dimensionless coefficients that are normalized by the scales of the pipe flow in the arms: $C_\Phi = 2\tau_\Phi / \rho a^3 U^2$ and $C_\tau = 2\tau / \rho a^3 U^2$. The gray diagonal line represents a perfect match of the theory with experiment, implying that the measured torque can be entirely accounted for by that associated with flux. The forward (crosses) and reverse (dots) cases are included for all 7 sprinkler designs, and the two modes are shown on different axis scales for clarity. The boxes reflect variability in the flow measurements that determine C_Φ and the torque measurements that yield C_τ. The forward mode of the overbend (yellow cross) is shown on the axes of the reverse mode, as appropriate given its torque magnitudes. The forward torque for the hookback is shown in absolute value to ease comparison with the forward cases. Across all geometries and both modes, the flux-based torque correctly accounts for the sign of the experimentally determined response. Moreover, the general trend across all measurements is well explained by the model as quantified by a correlation coefficient value of $R^2 = 0.99$. The theoretical values deviate on average from the measured torques by 5% and 30% for the forward and reverse cases, which are comparable to the variability in the measurements. Taken together, these findings indicate that flux-based torque is a primary contributor for all sprinkler geometries tested and for both modes of operation.

The flux theory explains the observed scales of the reverse and forward torques, specifically through the factor $\vec{r} \times \vec{u}$ that appears in the torque integral and relates to flow momentum density. In the reverse mode, this factor scales with the axial and transverse speeds as δU and $R_{\mathrm{IO}} V$ respectively, where δ is the millimeter-scale effective displacement of the internal jets due to their skewness and R_{IO} is the centimeter-scale distance to the inner orifice from the center of rotation. These two terms prove to contribute at the same order. In the forward mode for those designs with the outer orifice oriented azimuthally (i.e. not the straight nor overbend), the factor scales as $R_{\mathrm{OO}} U$, where R_{OO} is of order 10 cm. Hence $R_{\mathrm{OO}}/\delta \sim 100$ explains the forward versus reverse scales of torque in terms of the flow scales and differing orifice geometry. Moreover, that the forward torques are proportional to R_{OO} explains the ordering as standard, spiral, hookback, and overreach for increasing torque due to the increasing reach of their arms. The overbend is distinct in that its outer orifices are oriented radially and therefore its forward momentum density and torque scale with δU and $R_{\mathrm{OO}} V$, explaining why it responds in the forward mode with magnitudes typical of the reverse mode.

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The model framework is naturally extended to describe the dynamics of the free sprinkler by considering that the propulsive torque τ is resisted by hydrodynamic drag τ_D on the structure, thereby determining the steady-state rotational velocity Ω. Introducing the dimensionless drag coefficient $C_D = 2\tau_D/\rho a^5\Omega^2$ based on the expected quadratic dependence on rotation speed, then torque balance implies $C_\tau = C_D C_\Omega^2$ where $C_\Omega = a\Omega/U$. Hence $C_D = C_\tau/C_\Omega^2$ is a ratio of measurable quantities that provides a dimensionless metric reflecting the effectiveness with which torque is converted to speed. The plot of C_τ against C_Ω in Fig. 6*B* indicates that the inferred ordering of increasing drag goes as the standard and antichiral designs, spiral and hookback, and last overbend and overreach. The ranking is largely preserved between the two modes as expected if the drag values for the two rotation directions were comparable. Further, the results qualitatively match expectations that drag increases with the radial reach of the arms, and the standard and its antichiral twin prove to be optimal torque-to-speed converters among all designs tested.

The flux mechanism suggests a unifying picture of sprinklers in both modes and of different arm geometries. As shown in the *Top* panel of Fig. 6*C*, the forward sprinkler takes as input the isotropic diverging flow from the internal sink and ejects azimuthal jets from the curved arms, and the outbound flux of angular momentum powers its rotation. The reverse sprinkler shown in the *Lower* panel takes in isotropic converging flow from the far field and injects internal jets with slight skew and deflection, and the associated inbound flux propels the device. Both modes therefore involve fluidic mass flux due to source/sink flows that are transformed by the arm geometry to angular momentum flux which provides propulsive torque that is then converted into steady-state rotation by balancing drag.

Discussion and Conclusions

The experimental results presented here elevate the angular momentum flux hypothesis over other candidate descriptions for the operation of the reverse sprinkler. Across a variety of geometries, the observed sense of rotation and sign of propulsive torque is shown to be correlated with and explainable by the fluidic angular momentum entering the central hub. In contrast, geometries tailored to increase or decrease the total angular momentum of the fluid do not significantly affect the reverse response, and hence the notion that the steady-state rotation is determined by the balance of fluid and solid angular momentum lacks explanatory power. Similarly, designs that ought to enhance, cancel, or even invert the effects operating at the outer portions of the arms fail to generate the expected responses and hence such effects should be viewed as playing a secondary role. Internal flux, on the other hand, is shown by our measurements to serve as an order-one contributor to the reverse torque for all geometries tested. Moreover, this mechanism has the appealing aspect of unifying the reverse and forward cases, where flux in the latter case is external and readily associated with the jets issuing from the outer orifices. The nuanced reverse case involves internal jets that are subtly asymmetric due to transverse and skewed axial flows. These findings build on that of Wang, which reported motion and flow measurements for a single design similar to what we here call the standard (9). Our results are fully consistent while providing notable advances: 1) direct torque measurements, which were not presented in Wang nor elsewhere for any sprinkler design; 2) motion, flow, and torque measurements on a wide variety of sprinkler geometries; 3) generalization of the angular momentum flux model to include contributions from axial and transverse flow components; and 4) combination of all the above to show that the angular momentum flux theory applies generally for all geometries tested, both modes, and for Reynolds numbers of order 100 to 1,000.

Regarding the cause of the asymmetries in the internal jets, the leading hypothesis involves centrifugal effects acting on the flow rounding the bends in the arms, as originally raised by Beals (23) and supported by Wang (9). In the well-studied Dean problem for developed flow in a long coiled pipe (31, 32), inertial effects lead to dipolar vorticity in the pipe section and a velocity maximum displaced outwardly with respect to the bend, and these features are consistent with the transverse and axial flows documented here. Further, such effects are known to be significant for sufficiently high Dean number $\mathrm{De} = \mathrm{Re}\sqrt{a\kappa} \gtrsim 50$ where κ is the centerline curvature, and we estimate $\mathrm{De} \approx 200$ to 600 for our systems as based on the curvature of the bend nearest the hub. These correspondences speak to the plausible role of centrifugal effects even though the conditions do not meet the Dean assumptions of fully developed flow and uniform curvature. While the inner jet profiles depend somewhat on the flow path as set by the arm centerline shape, angular momentum flux is observed to be surprisingly robust. Greatly increasing the length over which the arms bend, as in the spiral and overbend geometries, does not markedly affect the internal momentum flux relative to the standard, nor does introducing an upstream counterbend as in the hookback design. A picture consistent with these results is that the final bend, whose shape is kept the same across all geometries (except straight), dominantly determines the reverse flux and torque. This bend is the closest to the internal orifice and has the maximal curvature tested here, and future work could assess the effect of varying how exactly the arms meet the hub.

The flux mechanism is perhaps the most parsimonious explanation, but it is not entirely distinct from other hypotheses. The idea of fluid action and solid reaction that is core to the total angular momentum picture is retained but recast in terms of fluidic angular momentum flux and the associated reactive torque on the structure. The failure of the former picture lies in the fact that the fluid–solid system is not closed, and the escape of angular momentum down the sink is in fact an essential feature of the steady state reverse mode. Regarding the hydrodynamic subtleties at the outer portions of the arms, these are critical to the flux picture and indeed cause the flow asymmetries in the jets that inject angular momentum into the hub. This raises questions for future work about how the associated torque manifests as pressure and shear stresses and where exactly on the surfaces of the solid their propulsive components are developed.

With a substantial set of experimental measurements now in hand, it seems the onus is on theory to provide quantitative understandings. The specific challenge posed by our work is to formally derive through control volume analyses how and why the internal flux dominates the reverse problem (33, 34). Here the presence of the sink seems to forbid the conventional approach of enclosing the solid and surrounding fluid together in a single control surface, since the system is not closed and momentum escapes. This difficulty does not seem to arise in the forward case, where the external flux is the main driver and no angular momentum enters from the isotropic source. A possible way forward is to consider nonsimply connected control regions with an interior surface enveloping the internal sink, in which case all flux and stress contributions must be accounted for over both

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control surfaces since no simplifications seem available a priori. Based on our findings, the target of such an analysis would be to explain if and why for the reverse case stress-flux terms over the exterior are subdominant to those in the interior.

Another major opportunity for future research involves direct numerical simulations, which have not been reported in the literature. In particular, flow–structure simulations that solve for the rigid-body rotational dynamics coupled to the fluid motions induced by an internal sink/source could provide the spatially and temporally resolved flow and force information needed for a rigorous and complete control volume analysis (35–37). The planar setting of 2D flow seems like a natural starting place, and special considerations include the boundary conditions and especially the sink/source region where the flows are predicted in the reverse case to be spatiotemporally complex yet critical to the operating mechanism. Given the results presented here, it may prove critical to capture 3D flow effects within the arms, and moreover sprinkler geometries such as the overbend design are inherently 3D. For future computational and theoretical studies generally, the wealth of experimental results presented here can guide the investigations, provide data needed for validation, and therefore contribute to building a consensus understanding of the reverse sprinkler. More broadly, this work connects to the class of applications involving forces and torques exerted on structures via the emission of fluid jets, as for propulsion or thrust vectoring (1–3, 38, 39). From this view, our findings demonstrate a form of geometric control via the shape of the conduit, for which the effects are especially subtle for inwardly aimed jets.

Materials and Methods

Here we provide additional details about the experimental setups, manufacturing procedures, calibrations, and data analysis methods.

Manufacturing. The sprinkler arms are constructed from 6-mm inner diameter acrylic plastic tubes. The various shapes are rendered by heat-forming using a heat gun and metallic panels that serve as molds against which the softened tubes are bent, pressed into place, and cooled to harden. Insertion of a foam core prevents collapse, and care must be taken to heat gently and evenly. The hub cylinders are clear acrylic stock tubing machined to length on a lathe (9). A bottom disk and top annular disk are laser cut and fused to the cylinders with acrylic solvent. Tight-tolerance holes are machined in the hub using a center finder and drill bit on a milling machine. The arms are inserted into these holes and fused with solvent. The helical stripe used for motion tracking is made by drawing a diagonal line on adhesive tape and affixing it around the hub cylinder.

Flow Control Apparatus. All three experimental setups shown in Fig. 2 are equipped with the apparatus shown in B. The flow is driven by a siphon with the volumetric rate Q set by the height differential of the water levels in the main and side tanks, both of which have holes for overspilling to a common reservoir. Depending on the mode, one tank or the other has higher level and is drained via the siphon and replenished by a pump which draws from the reservoir. The pump rate is set just greater than Q so as to minimize the induced flows in the tank, and Q is measured by intercepting the overspill from the lower tank.

Rotational Motion Measurements and Procedures. Low friction rotation is enabled by the capillary bearing system depicted in Fig. 2*A* and described previously (9). The centering or trapping of the sprinkler between the inner siphon and outer ring is due to surface tension repulsion of downward and upward menisci. The annular top surface of the sprinkler hub is made to sit slightly below the water surface by controlling the buoyancy via the size of an air bubble trapped inside the hub between the cylinders, and the glass siphon and ring are wetting. A typical experimental run begins by thoroughly cleaning, rinsing, and assembling all components of the bearing, flow control apparatus, and the sprinkler. Degassed and filtered water is used. Once slight negative buoyancy of the sprinkler is achieved, it is inserted into the bearing. A side-view camera (Nikon D610) captures photos of the helical stripe at a rate of 2 frames per minute. A custom code measures the height of the stripe along a vertical line through the middle of the hub, and these data are converted to the angle $\theta(t)$ via the known pitch of the helix. The experimental program entails letting the sprinkler come to rest after setup and then triggering the camera to record a 2-min quiescent period at $Q = 0$. The siphon valve is then abruptly opened to allow flow throughput at the prescribed Q for several minutes until steady state is reached, and finally another quiescent period is recorded before stopping the video. Three such trials are run for each geometry in each mode.

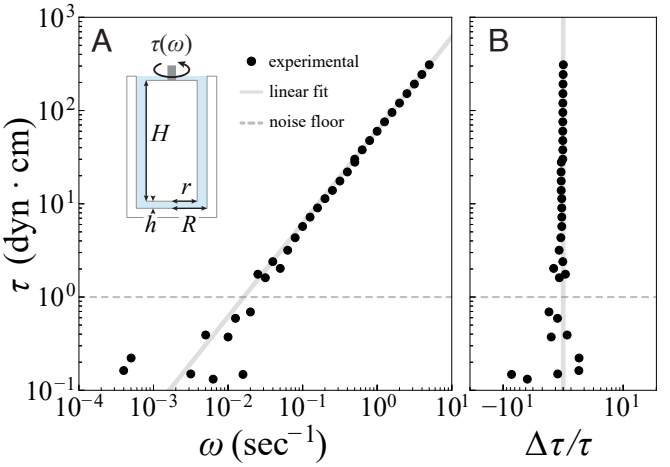


Fig. 7. Calibration of the rheometer as a torque gauge. (*A*) Measured steady-state torque as a function of rotation rate for the Taylor-Couette geometry of concentric cylinders with a fluid-filled gap. (*B*) Deviations from a linear fit identify the noise floor.

The bearing presumably induces friction whose magnitude is set by capillary interactions between the microscopically rough contact lines on the rotor (sprinkler) and stators. Attempts to measure or infer this resistance have proven difficult since it is small and substantially less than the typical propulsive torques for most conditions studied here. Namely, previous experiments show reverse rotations down to about Re $=$ 100, suggesting that friction is comparable to or less than the associated propulsive torque (9). Because the motion measurements reported here are for higher Re $=$ 500, and since propulsive torque scales quadratically with flow speed, we therefore infer that friction affects the dynamics at the order of a few percent. It would be desirable in a future work to measure the friction directly, for which the rheometer is insufficiently sensitive given its noise floor reported in what follows.

Torque Measurement Procedures and Rheometer Calibration. We employ the rheometer AR-G2 (TA Instruments Inc.) which uses compressed-air and magnetic bearings to levitate and rotate the spindle (measurement head) for low-friction movement and high-precision torque measurement. The device is used here as a torque gauge by operating it in a mode in which the moment required to keep the spindle stationary is reported as time-series data. This approach proved more reliable than a system involving mounting the sprinkler to a torsion spring and measuring rotations, which however could be improved and tested in the future. As shown in Fig. 2*B*, we fit the rheometer spindle with a 3D-printed mounting rig that secures to the sprinkler hub and allows the entry through its top of a siphon tube that may inject or withdraw fluid. A main tank sits on the base of the rheometer, and the sprinkler is submerged in water to its top. The height of the side tank is first adjusted to achieve the desired flow rate Q. We then institute a 10-min program throughout which the sprinkler is effectively static (i.e. set to the minimal speed of $\Omega = 10^{-9}$rad/s ≈ 0) while the torque τ is output at the maximal sample rate of 1 Hz. The system is left idle for 1 min with no flow input ($Q = 0$), then the flow is abruptly started by opening a valve on the siphon line, the prescribed Q is left on for 8 min, and then the valve is abruptly shut followed by another 1-min idle period. The data

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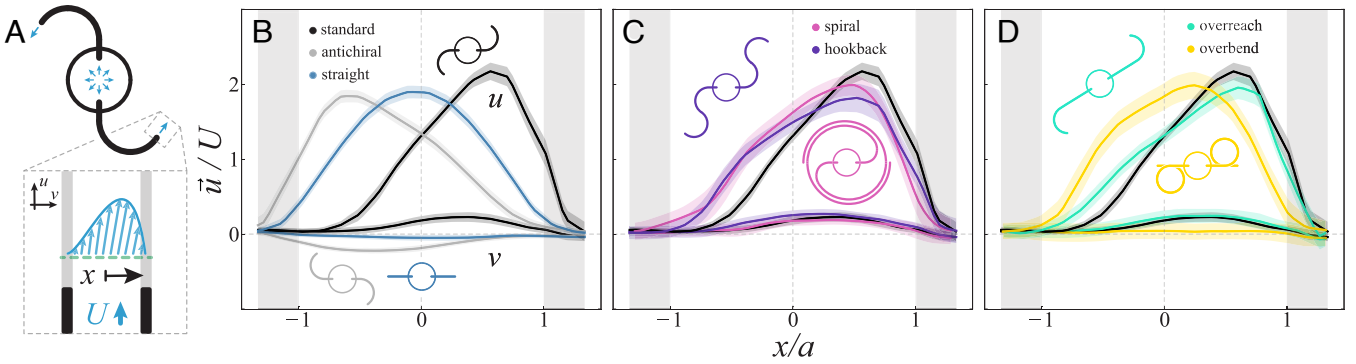


Fig. 8. Flow velocimetry at the outer orifices in the forward mode, where the presentation in panels *A*–*D* parallels that of Fig. 5.

during the flow-on period is processed by subtracting its mean by that of the flow-off periods. For a given sprinkler geometry and given Q, four such trials are run and their ensemble average and SDs constitute the $\tau(Q)$ markers and bars shown in plots.

The target range of Re from about a hundred to several thousands is set at the low end by the onset of directed reverse motion (9) and at the upper end by turbulence in the pipe arms. In practice, the reverse mode is limited on the low end by measurement variability, and hence we typically cover $\mathrm{Re} = [500, 1, 500]$ for high signal to noise. The noise floor of the rheometer is determined by conducting experiments using a Taylor-Couette geometry in which an inner cylinder is connected to the spindle and rotates at rate ω within a fixed outer cylindrical chamber, where the thin gap is filled with water. The data of Fig. 7*A* confirm the linear torque–speed relation expected at low Taylor numbers (40), and the deviations shown in B identify a noise floor of 1 dyn · cm. This dynamic calibration proved more reliable than a static one using a mass-pulley-string method, which incurs static friction in the pulley bearings. The static method did, however, yield the useful observation that the torque outputs lacked significant temporal fluctuations, which speaks to the stability of the rheometer's torque-motion feedback system. Hence we interpret the observed sprinkler torque variations as dominantly deriving from unsteady flows (9).

Flow Field Measurement via PIV. Measurement of the orifice flows is done with the sprinkler fixed to a glass-bottomed tank while using the flow control apparatus to set the mode and $\mathrm{Re} = 250$, this value being representative of higher flow rates for which the jet profiles show weak dependence on Re (9). The water is seeded with hollow glass microparticles (3M Inc., 50 μm diameter), and a light sheet formed by a cylindrical lens and continuous, high-intensity green laser (CNI Laser MGL-H-532-1W, 532 nm wavelength) is shone from the side, illuminating the horizontal plane at the center height of the arms. A high-speed camera (Vision Research Phantom V10) mounted vertically below the tank is used to capture 8-bit monochrome, 1,152 × 1,152 pixel, 400 frame-per-second videos of duration 2.5 s for which the particles measure about 1 pixel. Velocity fields are computed via the package PIVlab (26, 27). The interrogation window is set to be 64 × 64 pixels for the first pass, then 32 × 32 pixel, and finally 16 × 16. Fig. 8 shows the flow profiles developed at the outer orifices in the forward mode, and these data complement the reverse mode of Fig. 5. The axial components of the flow are skewed toward the outside of the bend in the arms and the transverse components are similarly signed. The transverse components are notably stronger in the forward mode than reverse, perhaps due to the jets being unconfined. For the geometries whose outer orifices are oriented azimuthally, the transverse component contributes no flux-based torque due to the factor $\vec{r} \times \vec{u}$ vanishing in the integral for Φ described in the main text. In contrast, the overbend and straight designs have transverse components that contribute to Φ at the 70% and 95% level.

Flow Field Interpolation. The integral expression for angular momentum flux requires the full 2D flow velocity field over the disk-shaped region spanning an orifice, whereas the PIV experiments provide a 1D transect (Fig. 5). We therefore adopt an interpolation procedure to infer the former from the latter, as was shown previously to have errors of about 5% (9). Expressing the 2D field in polar coordinates as $u_{2D}(r, \theta)$ for $r \in [0, a]$ and $\theta \in [0, 2\pi)$, the two halves of each experimentally measured axial flow profile correspond to $u_{2D}(r = x, \theta = 0) = u_{1D}(x \geq 0)$ and $u_{2D}(r = |x|, \theta = \pi) = u_{1D}(x < 0)$. Our interpolation scheme is based on analytical solutions to the Dean problem for flow in curved pipes: $u_{2D}(r, \theta) = A(r) + B(r)\cos(\theta) + \cdots$, where $A(r)$ and $B(r)$ are polynomials that depend on Re, diameter, and curvature in the context of fully developed flow in long, gently coiled tubes (31, 32, 41). We assume that the dominant shape modes of the 2D profile are captured by truncation of the series at the two terms shown, where $A(r)$ and $B(r)$ are unique to each arm geometry at a given Re. These are then determined from the PIV measurements by the 1D–2D matching conditions: $A(r) = [u_{1D}(x \geq 0) + u_{1D}(x < 0)]/2$ and $B(r) = [u_{1D}(x \geq 0) - u_{1D}(x < 0)]/2$. Examples of the resulting 2D axial flow profiles for the straight and standard geometries are presented in Fig. 9. For the transverse flow component, we assume the measured speed at each point x along the transect falls off in the vertical direction as a parabolic profile that goes through 0 at the walls: $v_{2D}(x, y) = v_{1D}(x)\left[1 - y^2/(a^2 - x^2)\right]$ for $x \in [-a, a]$ and $y \in [-\sqrt{a^2 - x^2}, \sqrt{a^2 - x^2}]$. The transverse components contribute to the reverse torque integral Φ at the 30% to 50% level across the geometries.

Flow Development Along the Overreach Arms. The long straight segment of each arm in the overreach design is expected in the reverse mode to partially symmetrize the distorted flow coming from the elbow before it reaches the inner orifice. Such effects are considerable when the pipe length is comparable to or longer than the characteristic length over which momentum diffuses across the pipe radius. The latter is referred to as the hydrodynamic development length scale $L_H = 2ak\mathrm{Re}$ and is well studied for the entrance flow into a pipe where boundary layers thicken to span the section (42–44). Here the constant $k = 0.05$ is set by the conventional definition that the flow develops to within 99% of its limiting profile. Applying the condition $L_H(\mathrm{Re}) = L = 10$ cm that the development length scale match the physical length of the straight segment yields a critical value of $\mathrm{Re} = 350$. We therefore conclude that development

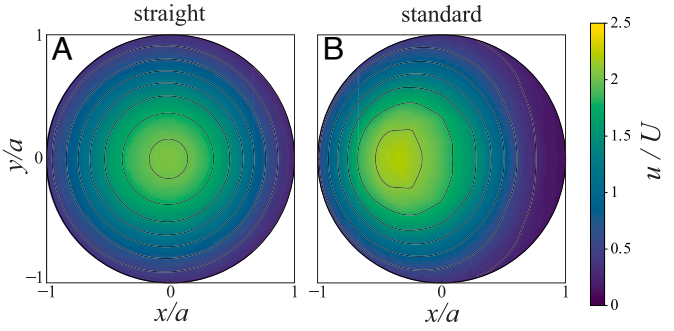


Fig. 9. Interpolated axial component of the flow velocity profile across the inner orifice for the reverse mode of the straight (*A*) and standard (*B*) geometries.

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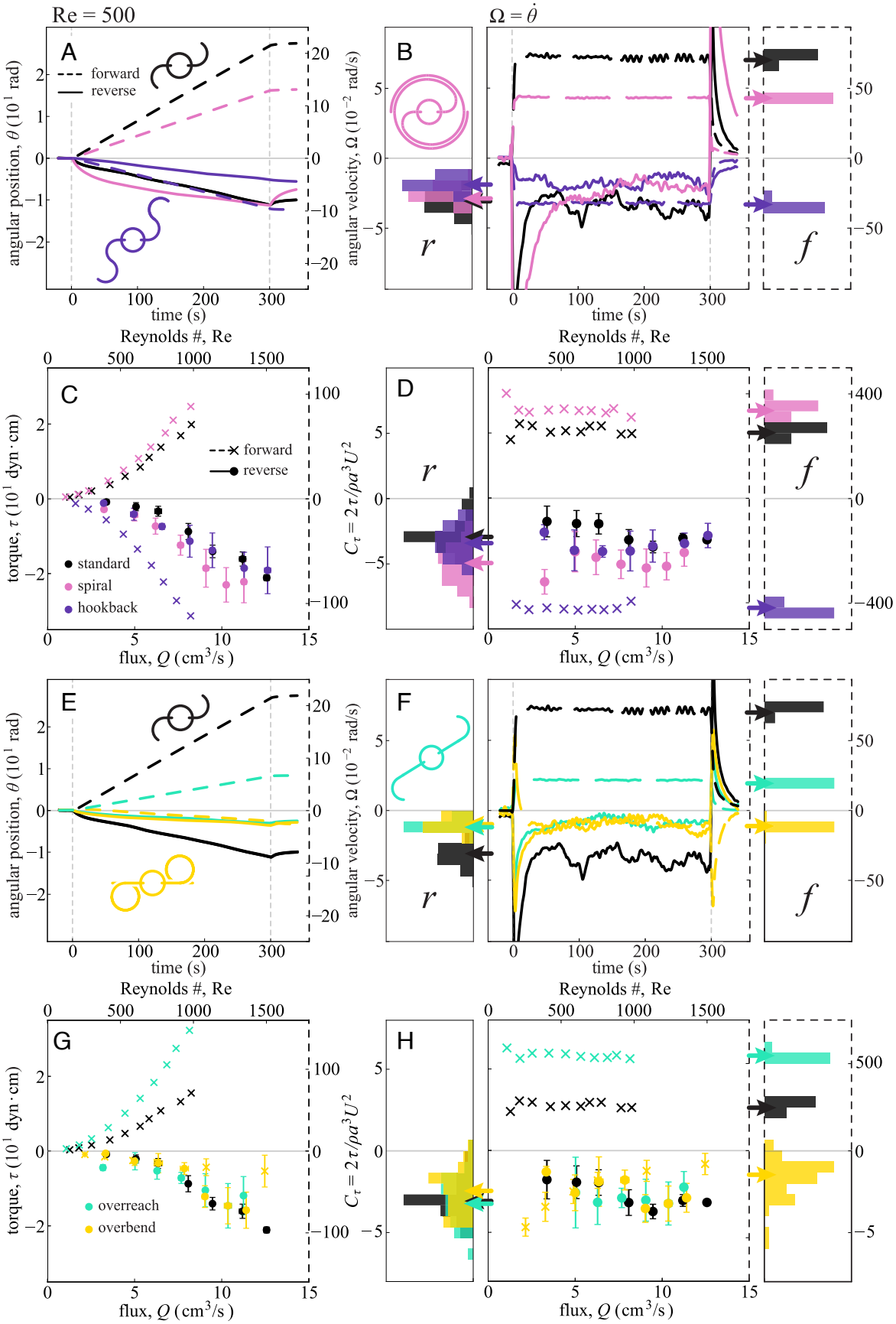


Fig. 10. Extended motion and torque datasets for the spiral, hookback, overreach, and overbend geometries, where the panels *A*–*H* follow that of Fig. 3.

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effects are important, as evidenced by the results of Fig. 5*D* showing the more symmetric profile for the overreach (green) as compared to the standard (black).

Flow development effects in the overreach impact the comparison of the torque and flow measurements since they are conducted at somewhat different conditions, i.e. Re $\in$ [500, 1,500] for the former (Fig. 10*H*) and Re $=$ 250 for the latter (Fig. 5*D*). Theoretical considerations allow the flux at the higher range of Re to be estimated from the measured flow and flux at lower Re. The flow profile just downstream of the bend for the overreach can be assumed similar to the measured profile at the inner orifice for the standard at the same Re. The flow thereafter develops exponentially in space along the long segment toward the measured profile at the inner orifice of the overreach. We may consider a quantity C_Φ that measures the deviation of the flow profile from its fully developed form, and this metric decays along the length x of the long segment of the overreach arm as $C_\Phi(x;\mathrm{Re}) = C_\Phi(0;\mathrm{Re})\exp(-x\ln 100/L_H)$. Here $\ln 100$ relates to the definition of L_H as 99% convergence. As the notation suggests, the coefficient of angular momentum flux C_Φ can serve as the deviation metric, i.e. it is expected to be nonzero for the skewed profile just after the elbow and then decay exponentially as the flow symmetrizes. Then $C_\Phi(0;\mathrm{Re}^*) = C_{\Phi,\mathrm{SS}}(\mathrm{Re}^*)$ is the flux measured from the standard sprinkler at $\mathrm{Re}^* = 250$, and $C_\Phi(L;\mathrm{Re}^*) = C_{\Phi,\mathrm{OR}}(\mathrm{Re}^*)$ is that measured from the overreach. Further, $C_{\Phi,\mathrm{SS}}(\mathrm{Re} > \mathrm{Re}^*) \approx C_{\Phi,\mathrm{SS}}(\mathrm{Re}^*)$ since it was shown previously that the standard sprinkler's flow profile and flux coefficient saturate at higher Re (9). Combining the above relations, the flux coefficient coming from the overreach at higher $\mathrm{Re} > \mathrm{Re}^*$ can be expressed in terms of coefficients measured at Re^*: $C_{\Phi,\mathrm{OR}}(\mathrm{Re}) = C_{\Phi,\mathrm{SS}}(\mathrm{Re}^*)\left[C_{\Phi,\mathrm{OR}}(\mathrm{Re}^*)/C_{\Phi,\mathrm{SS}}(\mathrm{Re}^*)\right]^{\mathrm{Re}^*/\mathrm{Re}}$. The value of C_Φ reported in Fig. 6*A* for the reverse mode of the overreach design is corrected according to this formula by averaging over the range Re $\in$ [500, 1,500] relevant to the rheometer measurements of C_τ.

Data, Materials, and Software Availability. All study data are included in the main text.

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1. S. L. Dixon, C. Hall, *Fluid Mechanics and Thermodynamics of Turbomachinery* (Butterworth-Heinemann, 2013).
2. J. Horlock, H. Marsh, Fluid mechanics of turbomachines: A review. *Int. J. Heat Fluid Flow* **3**, 3–11 (1982).
3. R. D. Sandberg, V. Michelassi, Fluid dynamics of axial turbomachinery: Blade-and stage-level simulations and models. *Annu. Rev. Fluid Mech.* **54**, 255–285 (2022).
4. E. H. Dowell, K. C. Hall, Modeling of fluid-structure interaction. *Annu. Rev. Fluid Mech.* **33**, 445–490 (2001).
5. M. J. Shelley, J. Zhang, Flapping and bending bodies interacting with fluid flows. *Annu. Rev. Fluid Mech.* **43**, 449–465 (2011).
6. T. Richter, *Fluid-Structure Interactions: Models, Analysis and Finite Elements* (Springer, 2017), vol. 118.
7. E. Mach, *The Science of Mechanics: A Critical and Historical Exposition of its Principles* (Open court publishing Company, 1893), pp. 299–302.
8. R. P. Feynman, R. Leighton, *"Surely You're Joking, Mr. Feynman!": Adventures of a Curious Character* (Random House, 1992).
9. K. Wang, B. Sprinkle, M. Zuo, L. Ristroph, Centrifugal flows drive reverse rotation of Feynman's sprinkler. *Phys. Rev. Lett.* **132**, 044003 (2024).
10. J. A. Wheeler, The young Feynman. *Phys. Today* **42**, 24–28 (1989).
11. E. Creutz, Feynman's reverse sprinkler. *Am. J. Phys.* **73**, 198–199 (2005).
12. A. T. Forrester, Inverse sprinklers: A lesson in the use of a conservation principle. *Am. J. Phys.* **54**, 798–799 (1986).
13. A. K. Schultz, Comment on the inverse sprinkler problem. *Am. J. Phys.* **55**, 488–488 (1987).
14. A. T. Forrester, Comments on a letter by Ak Schultz. *Am. J. Phys.* **55**, 488–489 (1987).
15. R. E. Berg, M. R. Collier, The Feynman inverse sprinkler problem: A demonstration and quantitative analysis. *Am. J. Phys.* **57**, 654–657 (1989).
16. E. R. Lindgren, The transport of momentum theorem. *Am. J. Phys.* **58**, 352–357 (1990).
17. S. Dasgupta, P. Mitra, S. Sengupta, More on inverse sprinklers. *Eur. J. Phys.* **11**, 311 (1990).
18. M. R. Collier, R. E. Berg, R. A. Ferrell, The Feynman inverse sprinkler problem: A detailed kinematic study. *Am. J. Phys.* **59**, 349–355 (1991).
19. A. Jenkins, An elementary treatment of the reverse sprinkler. *Am. J. Phys.* **72**, 1276–1282 (2004).
20. M. P. Païdoussis, M. Tétreault-Friend, Aspirating cantilevers and reverse sprinklers. *Am. J. Phys.* **77**, 349–353 (2009).
21. A. Jenkins, Sprinkler head revisited: Momentum, forces, and flows in machian propulsion. *Eur. J. Phys.* **32**, 1213 (2011).
22. W. Rueckner, The puzzle of the steady-state rotation of a reverse sprinkler. *Am. J. Phys.* **83**, 296–304 (2015).
23. J. Beals, New angles on the reverse sprinkler: Reconciling theory and experiment. *Am. J. Phys.* **85**, 166–172 (2017).
24. Q. M. Nguyen, J. Abouezzi, L. Ristroph, Early turbulence and pulsatile flows enhance diodicity of tesla's macrofluidic valve. *Nat. Commun.* **12**, 2884 (2021).
25. Q. M. Nguyen *et al*., Flow rectification in loopy network models of bird lungs. *Phys. Rev. Lett.* **126**, 114501 (2021).
26. W. Thielicke, "The flapping flight of birds: Analysis and application," PhD thesis, Rijksuniversiteit Groningen, Groningen, The Netherlands (2014).
27. E. Stamhuis, W. Thielicke, Pivlab-towards user-friendly, affordable and accurate digital particle image velocimetry in matlab. *J. Open Res. Softw.* **2**, 30 (2014).
28. W. Thielicke, R. Sonntag, Particle image velocimetry for matlab: Accuracy and enhanced algorithms in pivlab. *J. Open Res. Softw.* **9**, e334 (2021).
29. G. Stokes, On the theories of the internal friction of fluids in motion, and of the equilibrium and motion of pendulums. *Trans. Camb. Philos. Soc* **8**, 67 (1845).
30. K. Tsiolkovsky, *Reactive Flying Machines* (Izdatelstvo Akademii Nauk SSSR, Moscow, Russia, 1954).
31. W. Dean, XVI. Note on the motion of fluid in a curved pipe. *Lond. Edinb. Dublin Philos. Mag. J. Sci.* **4**, 208–223 (1927).
32. W. Dean, J. Hurst, Note on the motion of fluid in a curved pipe. *Mathematika* **6**, 77–85 (1959).
33. O. Reynolds, *The Sub-Mechanics of the Universe* (University Press, Cambridge, 1903), vol. 3, p. 9.
34. L. Prandtl, O. G. Tietjens, Fundamentals of hydro-and aeromechanics (Dover, 1934).
35. B. Kallemov, A. Bhalla, B. Griffith, A. Donev, An immersed boundary method for rigid bodies. *Commun. Appl. Math. Comput. Sci.* **11**, 79–141 (2016).
36. Y. Kim, C. S. Peskin, A penalty immersed boundary method for a rigid body in fluid. *Phys. Fluids* **28**, 033603 (2016).
37. N. Nangia, H. Johansen, N. A. Patankar, A. P. S. Bhalla, A moving control volume approach to computing hydrodynamic forces and torques on immersed bodies. *J. Comput. Phys.* **347**, 437–462 (2017).
38. R. D. Flack, *Fundamentals of Jet Propulsion with Applications* (Cambridge University Press, 2005), vol. 17.
39. F. Xue, G. Yunsong, Y. Wang, H. Qin, Research on control effectiveness of fluidic thrust vectoring. *Sci. Prog.* **104**, 0036850421998137 (2021).
40. G. I. Taylor, VIII. Stability of a viscous liquid contained between two rotating cylinders. *Philos. Trans. R. Soc. Lond. Ser. A Contain. Pap. Math. Phys. Character* **223**, 289–343 (1923).
41. W. Dean, LXXII. The stream-line motion of fluid in a curved pipe (second paper). *Lond. Edinb. Dublin Philos. Mag. J. Sci.* **5**, 673–695 (1928).
42. E. Sparrow, S. Lin, T. Lundgren, Flow development in the hydrodynamic entrance region of tubes and ducts. *Phys. Fluids* **7**, 338–347 (1964).
43. D. Fargie, B. Martin, Developing laminar flow in a pipe of circular cross-section. *Proc. R. Soc. Lond. A Math. Phys. Sci.* **321**, 461–476 (1971).
44. O. Pomerenk, S. C. Segura, F. Cao, J. Wu, L. Ristroph, Hydrodynamics of finite-length pipes at intermediate Reynolds numbers. *J. Fluid Mech.* **959**, A28 (2023).

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